# Emil Kasper

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## Experience

**Agile Robots SE**, Machine Learning Engineer (Working Student) — Munich, Germany
- Designing and running machine-learning training environments. — Oct 2026 – present
- Independently researching and implementing human-in-the-loop reinforcement learning (HIL-RL) for dual-arm manipulation tasks. — 2 months

**IVIA Lab, ETH Zurich**, Research Fellow (ETH SSRF) — Zurich, Switzerland
Supervised by Prof. Menna El-Assady. — July 2026 – Aug 2026
- Selected as one of 20 students from over 4,000 applicants for the ETH Student Summer Research Fellowship. — 2 months
- Working on agentic AI and agent-agent communication.

**Siemens**, Software Engineer (Working Student) — Vienna, Austria
- Contributing to a test simulation framework for the European Train Control System (ETCS). — July 2025 – June 2026
- Implementing the autonomous train operation (ATO) mode in C# for test simulations. — 1 year

**Institute for Visual Computing and Human-Centered Technology, TU Wien**, Teaching Assistant — Vienna, Austria
Course: Interface and Interaction Design. — Mar 2025 – June 2026 — 1 year 4 months
- Guiding students through the lifecycle of user-centered product development.
- Providing feedback and grading for 45 students.

**Machine Learning Research Unit, TU Wien**, Undergraduate Researcher — Vienna, Austria
Supervised by Prof. Clemens Heitzinger. — Aug 2025 – Apr 2026 — 9 months
- Proving better convergence and stability guarantees for Q-Learning and Speedy Q-Learning through batch sampling.
- Experimenting with PPO for robotic hand manipulation.

## Education

**MSc** — **ETH Zurich**, Computer Science — Zurich, Switzerland
- Major in Machine Intelligence. — Sept 2026 – June 2028

**BSc (Honors)** — **TU Wien (Vienna University of Technology)**, Computer Science — Vienna, Austria
- Four-year honors degree with a minor in machine learning. — Oct 2022 – May 2026
- Graduated with distinction.
- Received merit-based scholarships.

## Research

**Asking Your Advisor: Learning When to Consult Stronger Models in Agentic Systems**
Submitted to ICLR 2027. We train a meta-policy that learns when to consult a stronger advisor model and when to continue with a cheaper one, reaching a better success-cost Pareto frontier than standard advisor-executor setups. Preprint on arXiv.

## Student Engagements

**TU Wien Robotics Club**, Machine Learning Engineer — Vienna, Austria
- Implementing VLAs for robotic applications with an SO101 arm. — Apr 2026 – present
- Building a chess robot that autonomously plays chess and moves the pieces.

**European Student Robotics Association (ESRA)**, Executive Board Member, Partnerships — Europe
- Serving on the executive board of ESRA, the European Student Robotics Association. — Sept 2026 – present
- Driving partnerships to push the boundaries of European robotics.

**TU Wien Space Team**, Frontend Developer — Vienna, Austria
- Contributing to the "Lamarr" project at the 2025 European Rocketry Challenge (EuRoC). — Mar 2025 – Oct 2025
- Building frontend tooling for rocket system control.

## Programs

**TU Wien Informatics**, Bachelor with Honors — Vienna, Austria
- Selected for the Informatics excellence program, a performance-based honors track comparable to the Anglo-American honors model. — Oct 2024 – May 2026
- Performance-based admission and graduation places graduates among the top 5 percent of students at TU Wien.
- Extends the studies by one year with research-oriented and master's-level courses (program website).

**European Student Assembly 2026, EUC Voices**, Delegate — Strasbourg, France
- Representing the EULiST alliance at the European Student Assembly 2026. — Jan 2026 – Apr 2026
- Debating key issues and advocating for policy changes at the European level.
- Writing policy recommendations on empowering inclusive digital literacy and building resilience.

## Languages

**German**
Native

**English**
Fluent

**Russian**
Fluent